



Name: Cohort/Batch: Date: Score:

LEADER ELECTION PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Distributed Systems

TOTAL: 10 QUESTIONS

Q1 What is Leader Election and what problem in Distributed Systems does it solve?

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Q6 How does Leader Election handle reliability and high availability?

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Q2 What are the core components or concepts in Leader Election?

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Q7 What observability does Leader Election provide out of the box?

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Q3 How does Leader Election compare to its main alternatives?

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Q8 What are common performance bottlenecks in Leader Election?

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Q4 How do you get started with Leader Election for a new project?

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Q9 What security best practices apply when running Leader Election?

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Q5 What configuration options most affect Leader Election's behavior in production?

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Q10 What mistakes do teams commonly make when adopting Leader Election?

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ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Leader Election is a tool or platform

Mitigation

Leader Election is a tool or platform that automates or simplifies a specific concern in the Distributed Systems domain, reducing manual effort and operational complexity for engineering teams.

A6 Leader Election provides HA through leader election

Observability

Leader Election provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Leader Election has key building blocks that

Primitives

Leader Election has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Leader Election exposes metrics (Prometheus endpoint)

Information

Leader Election exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 Leader Election trades off simplicity against feature

Hardening

Leader Election trades off simplicity against feature richness differently than its alternatives. Choose Leader Election when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfiguration

Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install Leader Election via its package manager

Gotchas

Install Leader Election via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Leader Election with the principle of

Assessment

Run Leader Election with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency

Configuration

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and

Organization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.